Lorenzo Dusty Costa

FREELANCE DATA SCIENTIST

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Skills

Languages
Python, SQL, R, Java, Node.js, MATLAB

Data & ML
Machine learning, Neural networks, PyTorch, scikit-learn, XGBoost, MLflow, Quantile regression, Forecasting, Time series, Anomaly detection, Feature engineering, Polars

Mathematics
Probability, Monte Carlo, Mathematical modeling

Databases
PostgreSQL, MongoDB, BigQuery

Cloud & DevOps
GCP, AWS, Docker, Kubernetes, Git

Education

MSc in Mathematics
Università degli Studi di Milano
2011 – 2013 · 110/110 cum laude

BSc in Mathematics
Università degli Studi di Milano
2007 – 2011

Summer School in Industrial Mathematics
ECMI
Jul 2011

Certificates

Intro to SQL & Advanced SQL
Kaggle
2023

Machine Learning
Coursera
2018

Business Data Analysis
Talent Garden
2017

Languages

Italian Native

English Professional

Turkish Learning

Profile

Freelance data scientist with over 10 years of experience and a background in applied mathematics. I build predictive models, probabilistic forecasts and KPI pipelines, from problem definition to cloud deployment. I have worked across sectors including sport-tech, renewable energy, telco and retail consulting, and gaming. I work with SMEs, startups and consulting firms on proofs of concept, new builds, and the optimisation and maintenance of existing models and pipelines.

Experience

Freelance Data Scientist

May 2025 – Present

- End-to-end PoC on GCP for 5 wind farms: data ingestion, feature engineering, quantile XGBoost models with conformal calibration (CQR). nMAE ~10% of installed capacity on P50, on par with two commercial providers.
- With the client's team, built a new Python/Polars framework for post-match KPIs: reprocessing a full match went from over 90 minutes to a few seconds (-99%).
- Refactored Java and Python microservices and the PostgreSQL schema to switch to a new data provider, keeping compatibility with the previous one.

Lead Data Scientist · Wallabies

Nov 2023 – Apr 2025

- PyTorch neural networks predicting each team's goal distribution: +3–8 percentage points of 1X2 accuracy, using one model per league and new features encoding recent form.
- Python post-processing pipeline for a Serie A club's scouting: time per shortlist cut from half a day to just over an hour (-70%).

Data Scientist & Analytics Manager · Math&Sport

Jan 2020 – Oct 2023

- Tactical and physical KPIs (Python, scikit-learn) on tracking data up to 50 Hz for the real-time system used by Serie A coaching staff, delivered on tablets as numbers, charts and text.
- End-to-end development of the APIs (Node.js, MongoDB) that fed live data to Rai for the 2020 and 2021 Coppa Italia and Supercoppa finals.
- Automated the weekend setup on AWS (configuration, EC2 instances, data storage): from about 150 minutes per match to almost zero.
- Led a team of 4–5 people (data scientist and mobile developer); co-supervised two master's theses and co-authored a peer-reviewed paper.

Data Scientist · Spindox

Sep 2018 – Dec 2019

- Anomaly detection on time series of server hardware metrics for a telco client (Seasonal Hybrid ESD, Python and R), released as a service integrated into a monitoring dashboard.
- Model-based random forest (R, mobForest) estimating each booking's no-show probability: -15% no-shows according to client data.

Game Designer · Sisal
Jan 2014 – Aug 2018

- Odds models for sports betting: Markov chains for tennis, adapted to volleyball, and Poisson models for hockey and football.
- Mathematics of SuperEnalotto, Eurojackpot, SiVinceTutto and VinciCasa: prize structures, probabilities, Monte Carlo simulations and risk analysis (MATLAB, R). The 2016 SuperEnalotto formula produced 6 of the 10 largest jackpots in the game's history.

Publications

Using Temporal Convolutional Networks to estimate ball possession in soccer games
Expert Systems with Applications (Elsevier) · 2023
With Matteo Borghesi, Lia Morra, Fabrizio Lamberti

Projects

CopaMundial Website · torneocopamundial.it
Website to manage an amateur football tournament, built and deployed on free-tier services.

I authorise the processing of the personal data contained in this CV in accordance with Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003.